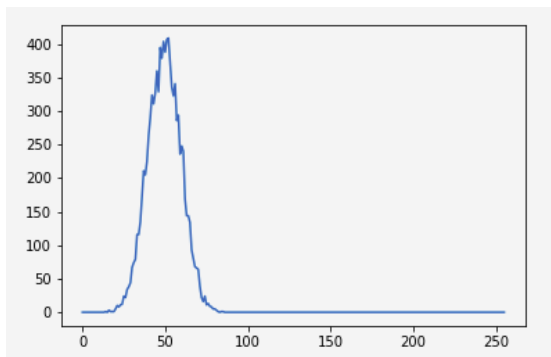


Filtering and Noise

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Department of Computer Science
The University of Sheffield



- The image is dark
 - The image is bright
 - The image makes use of a vary large dynamic range
-
- What is true about histogram equalization?
 - It's an automatic way to improve contrast
 - It always produces images that look "natural"
 - It changes the intensities of the image so that (approximately) all intensities are equally represented

Outline

- Images as Functions
- Filtering
- Linear Filters
- Non-linear Filters
- Sharpening Filters
- Padding

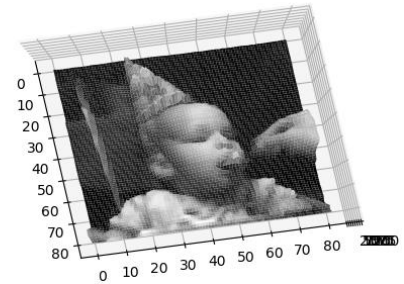
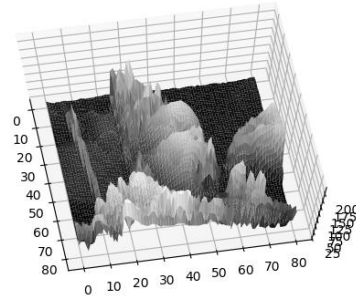
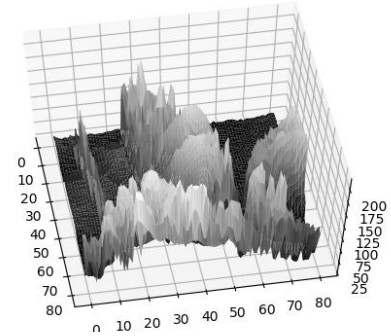
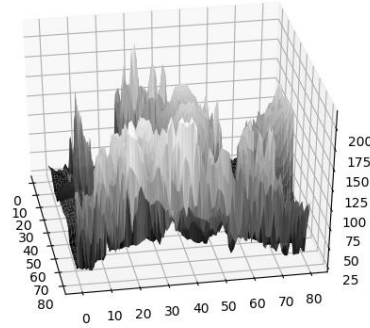
Image As Functions

- An image can be seen as a $\mathbb{R}^2 \rightarrow \mathbb{R}$ function: $I(x, y)$
 - that returns an **intensity** at position (x,y)
- For color images, we will have

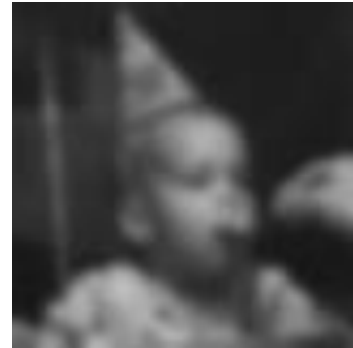
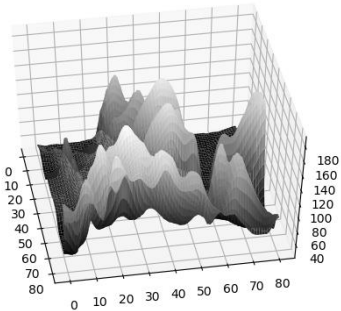
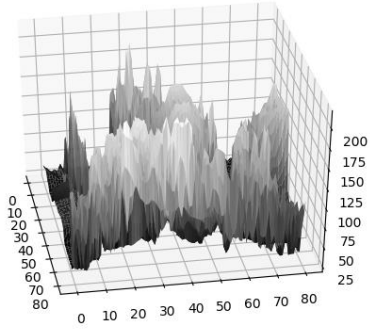
$$I : \mathbb{R}^2 \rightarrow \mathbb{R}^C$$

- where C is the number of channels

Image As A Function



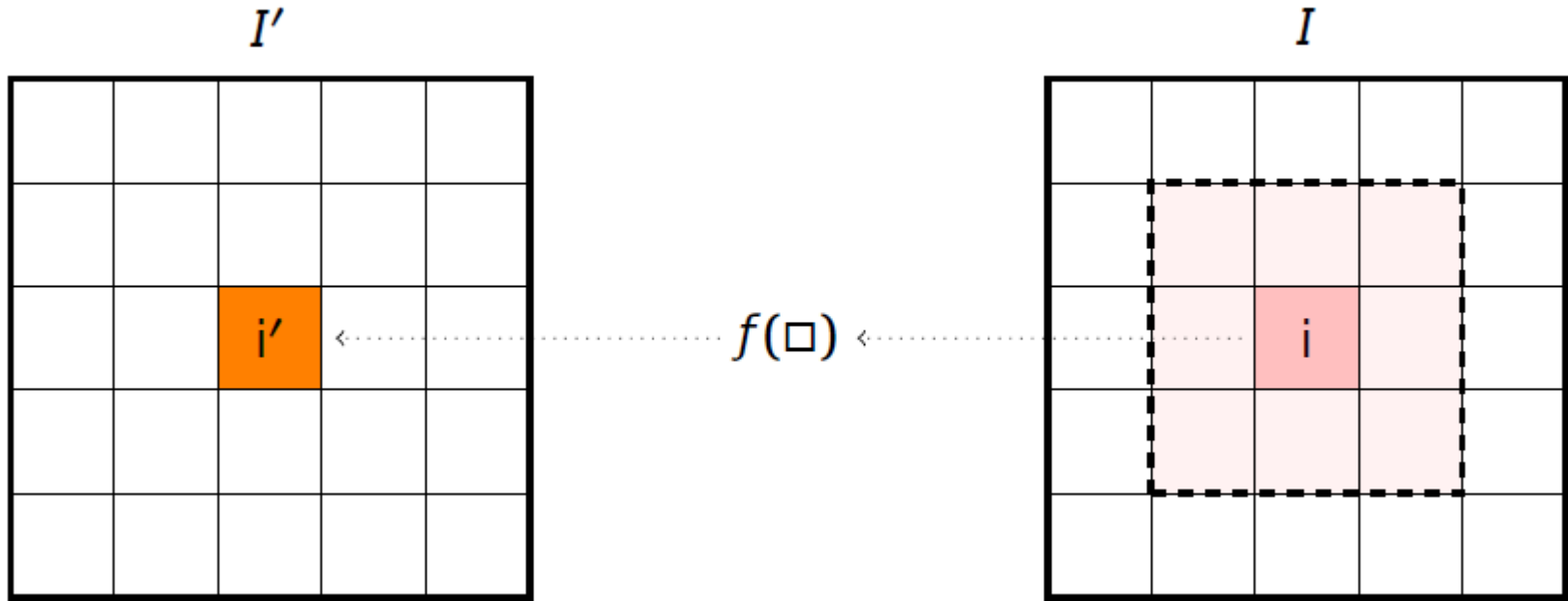
Example: Moving Average of the Function



Filtering

- Smoothing/blurring
- Noise reduction
- Sharpening/de-blurring
- Edge detection

Point Operators vs Filtering



Linear Filtering

- Cross-correlation
 - Measure of similarity of two signals
 - The response is maximum when they match exactly
- Convolution
 - “Measure” of the effect of one signal onto another
 - A kind of “sieve”

Cross-correlation

- If k is the radius (i.e. diameter/2) of the kernel h , then:

$$G[r, c] = \sum_{u=-k}^k \sum_{v=-k}^k h[u, v] F[r + u, c + v]$$

- Also written as: $G = h \star F$
- Kernels should be odd sized
 - The kernel is centered at (0,0), however, in programming (0,0) is top left corner

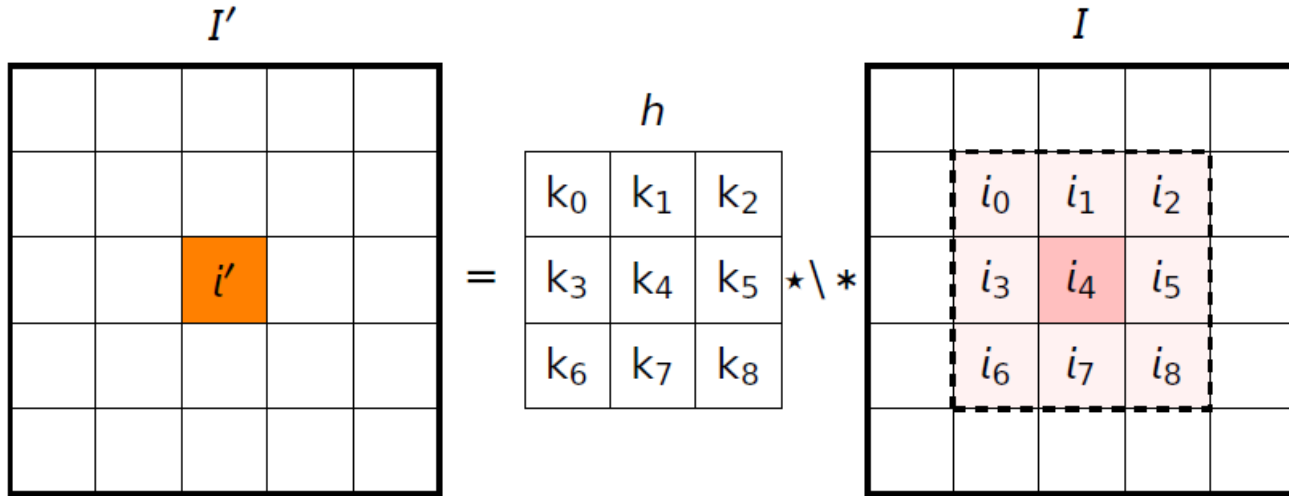
Convolution

- If k is the radius (i.e. diameter/2) of the kernel h , then:

$$G[r, c] = \sum_{u=-k}^k \sum_{v=-k}^k h[u, v] F[r-u, c-v]$$

- Also written as: $G = h * F$
- Kernels should be odd sized
 - The kernel is centered at (0,0), however, in programming (0,0) is top left corner

Cross-correlation vs Convolution



$$h \star I(2, 2) = k_0 i_0 + k_1 i_1 + k_2 i_2 + k_3 i_3 + k_4 i_4 + k_5 i_5 + k_6 i_6 + k_7 i_7 + k_8 i_8$$

$$h * I(2, 2) = k_8 i_0 + k_7 i_1 + k_6 i_2 + k_5 i_3 + k_4 i_4 + k_3 i_5 + k_2 i_6 + k_1 i_7 + k_0 i_8$$

Properties of Convolution

- Linear
- Commutative: $G * I = I * G$
- Associative: $(G * H) * I = G * (H * I)$
- It has an Unit Impulse: $U * I = I$
- Differentiation: $\frac{\partial}{\partial x}(H * I) = \frac{\partial H}{\partial x} * I$

Review

- Cross-correlation is maximum when both patches exactly match
- For circularly symmetric filters, convolution $\equiv$ cross-correlation
 - If $k_0 = k_8$ & $k_1 = k_7$ & $k_6 = k_2$ & $k_5 = k_3$

k_0	k_1	k_2
k_3	k_4	k_5
k_6	k_7	k_8

$$h \star I(2, 2) = k_0 i_0 + k_1 i_1 + k_2 i_2 + k_3 i_3 + k_4 i_4 + k_5 i_5 + k_6 i_6 + k_7 i_7 + k_8 i_8$$

$$h * I(2, 2) = k_8 i_0 + k_7 i_1 + k_6 i_2 + k_5 i_3 + k_4 i_4 + k_3 i_5 + k_2 i_6 + k_1 i_7 + k_0 i_8$$

Impulse Kernel

?

=

0	0	0
0	1	0
0	0	0

*



$$h * I(2, 2) = k_8 i_0 + k_7 i_1 + k_6 i_2 + k_5 i_3 + k_4 i_4 + k_3 i_5 + k_2 i_6 + k_1 i_7 + k_0 i_8$$

Impulse Kernel



=

0	0	0
0	1	0
0	0	0

*



Shift Kernel

?

=

1	0	0
0	0	0
0	0	0

*



$$h * I(2, 2) = k_8 i_0 + k_7 i_1 + k_6 i_2 + k_5 i_3 + k_4 i_4 + k_3 i_5 + k_2 i_6 + k_1 i_7 + k_0 i_8$$

Shift Kernel



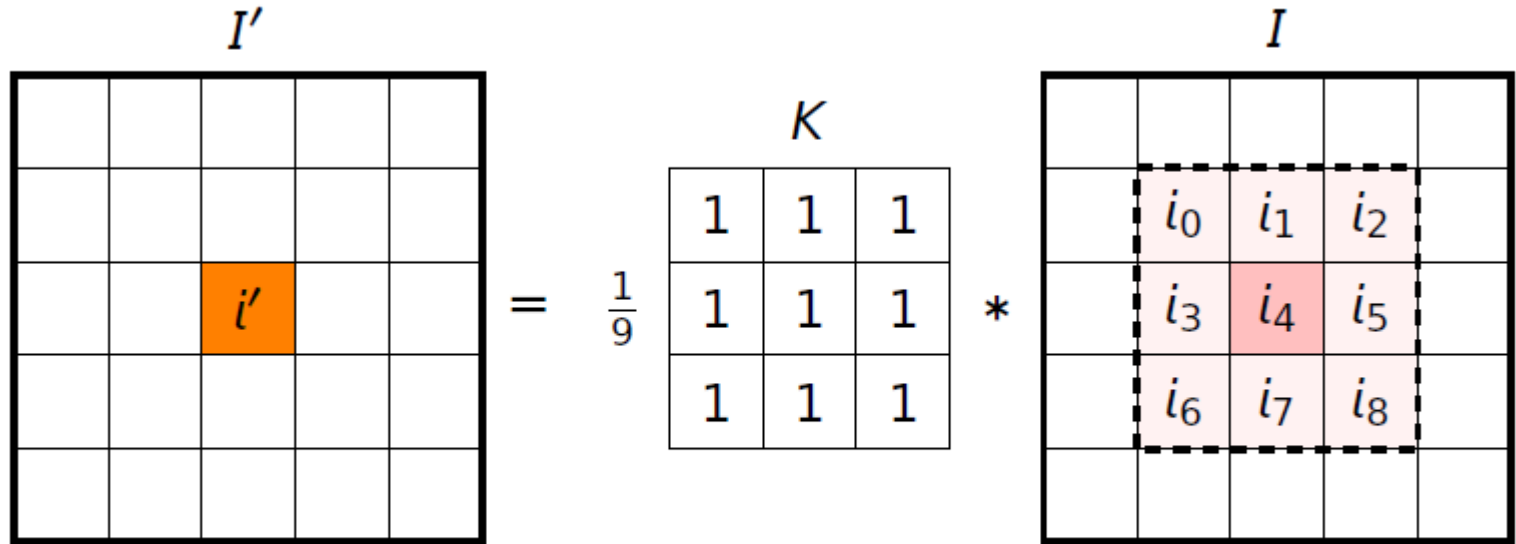
=

1	0	0
0	0	0
0	0	0

*



3 x 3 Average Kernel (Mean Filter)



$$i' = \frac{1}{9}(i_0 + i_1 + i_2 + i_3 + i_4 + i_5 + i_6 + i_7 + i_8)$$

3 x 3 Average Kernel (Example)

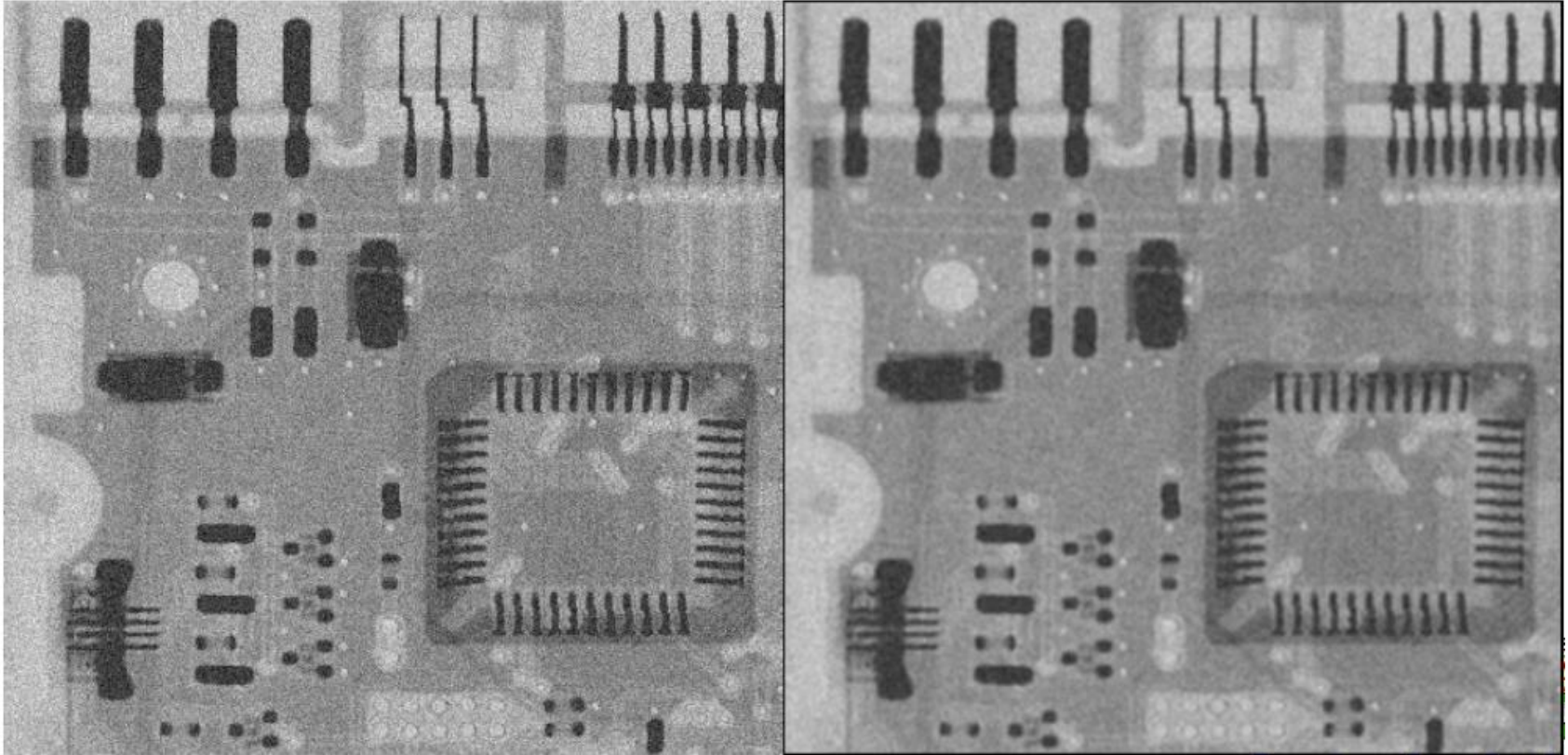
$$I' = \frac{1}{9} K * I$$

	100	100	100	
	100	100	100	
	100	100	100	

1	1	1
1	1	1
1	1	1

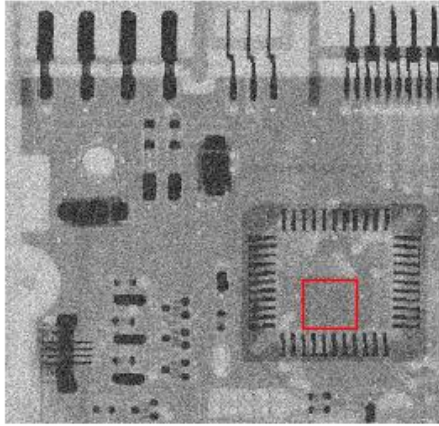
100	101	99	101	103
101	100	102	99	101
102	100	98	100	100
100	99	102	100	100
99	100	103	99	101

3 x 3 Mean/Average Smoothing

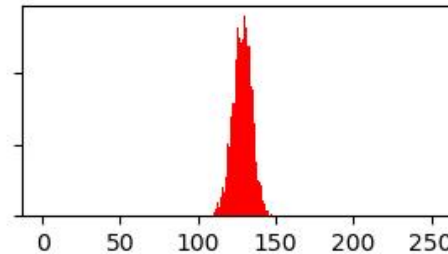
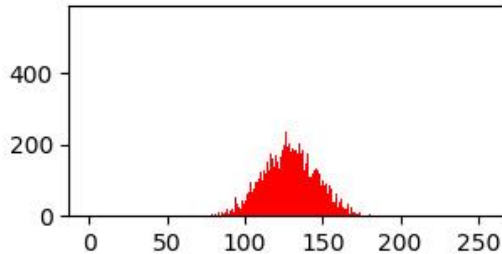
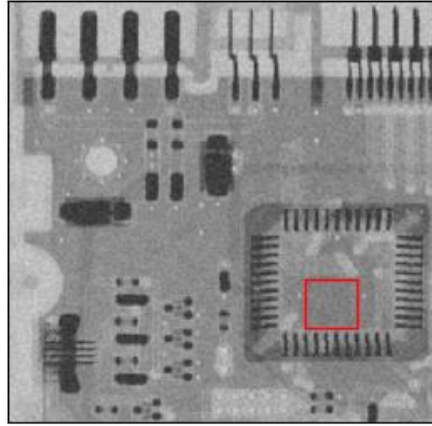


Noise: “Looking” At A “Constant” Area

Noisy



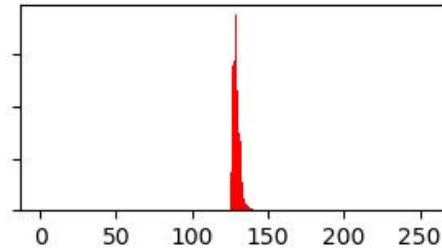
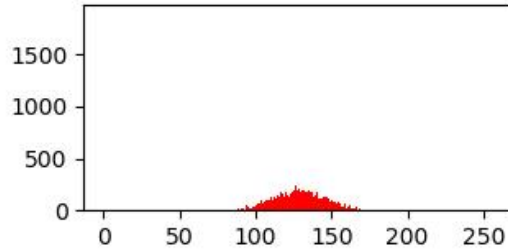
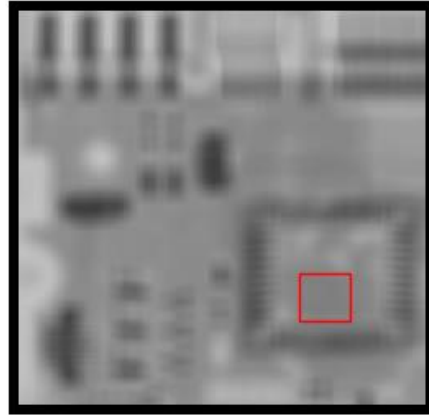
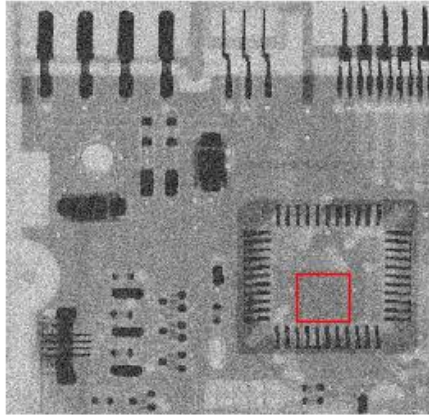
Mean filtered



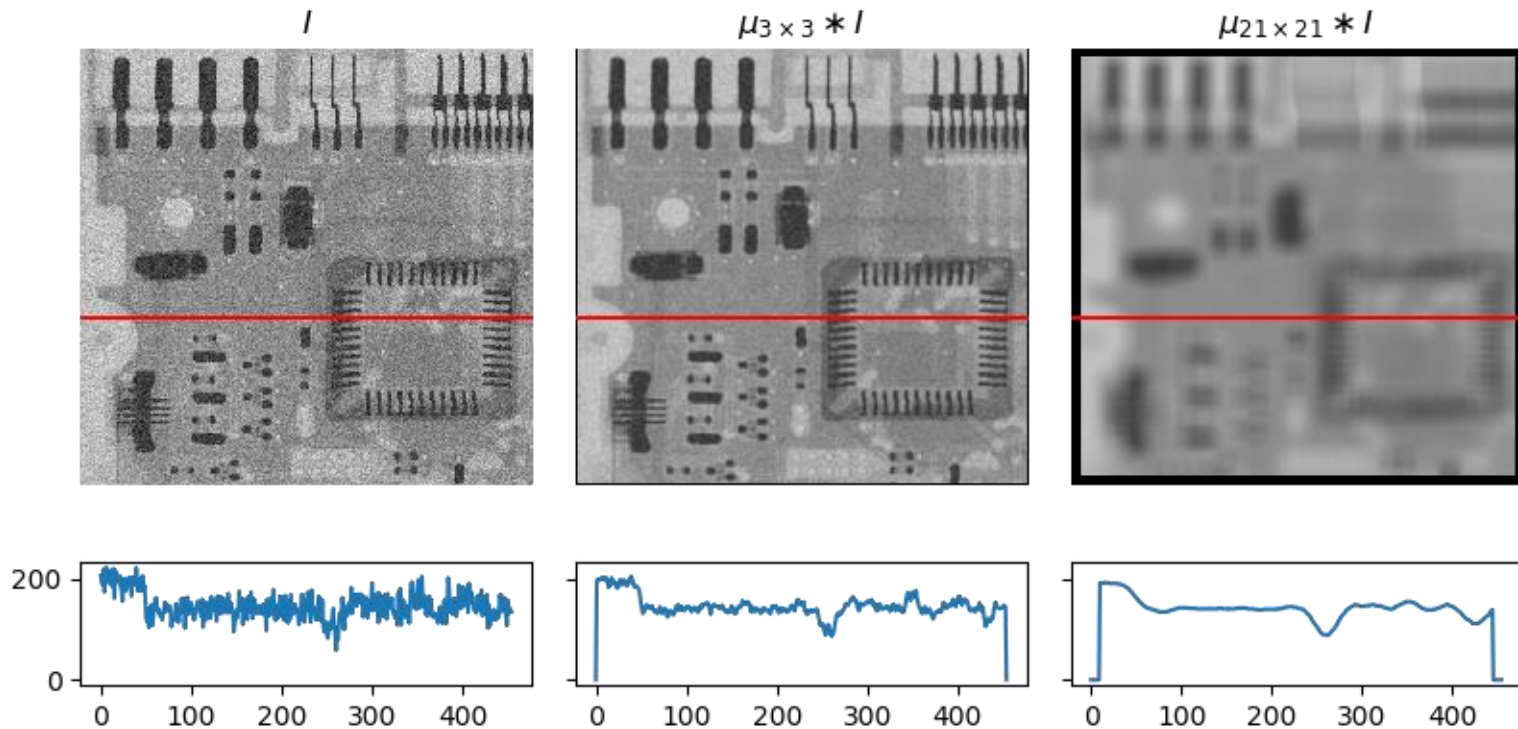
$$\frac{1}{9}$$

1	1	1
1	1	1
1	1	1

21 x 21 Mean/Average Smoothing



“Looking” At The Intensity Profile

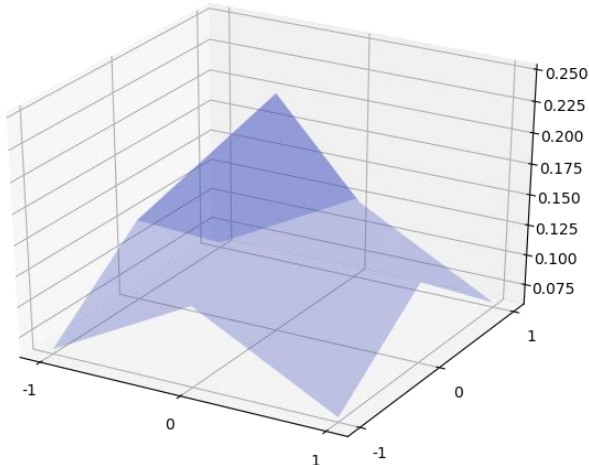


Weighted Average (Gaussian Filter)

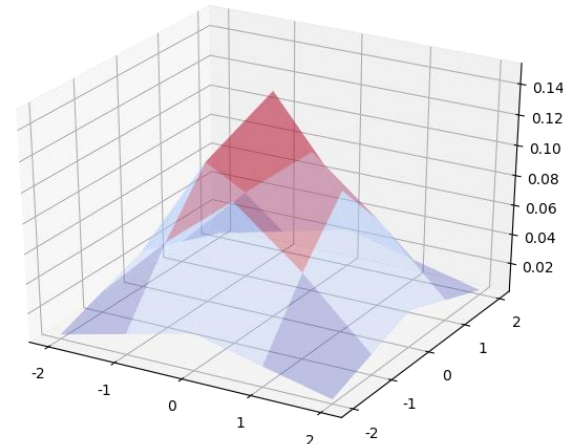
Given more importance to the central pixel (and closest to the center)

$$\frac{1}{16}$$

1	2	1
2	4	2
1	2	1


$$\frac{1}{273}$$

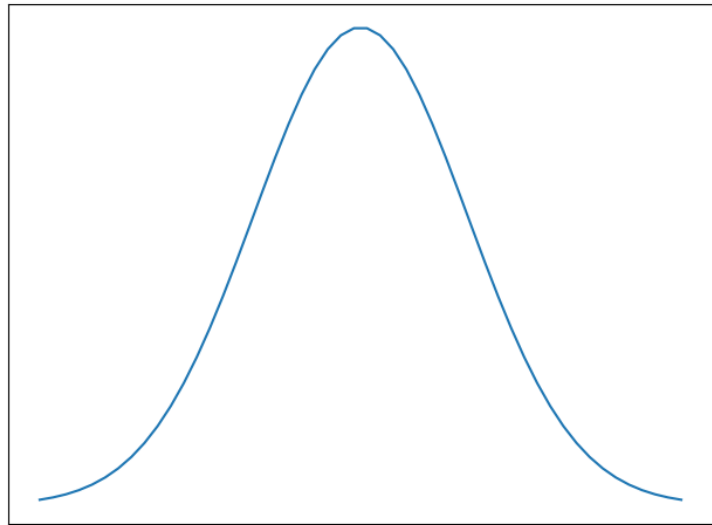
1	4	7	4	1
4	16	26	16	4
7	26	41	26	7
4	16	26	16	4
1	4	7	4	1



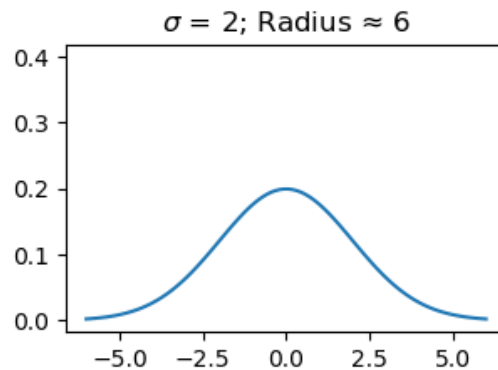
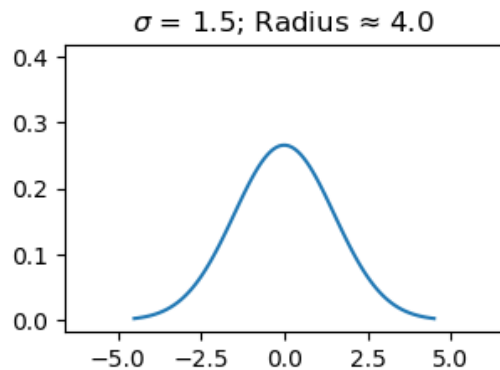
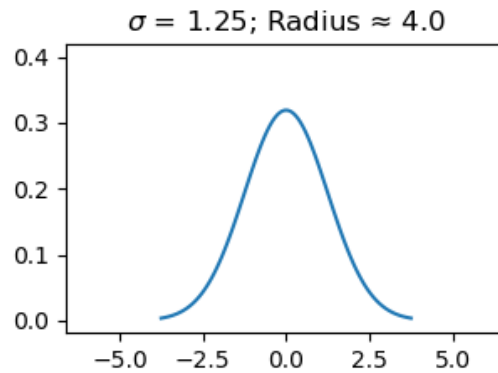
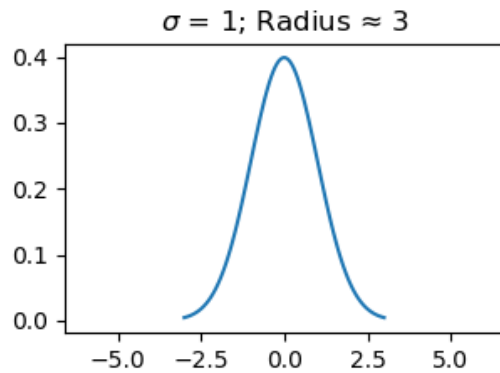
Gaussian Distribution – 1D

- Very important in statistics
- AKA “bell curve” (informally)
- AKA “normal distribution”: $\mathcal{N}(\mu, \sigma^2)$

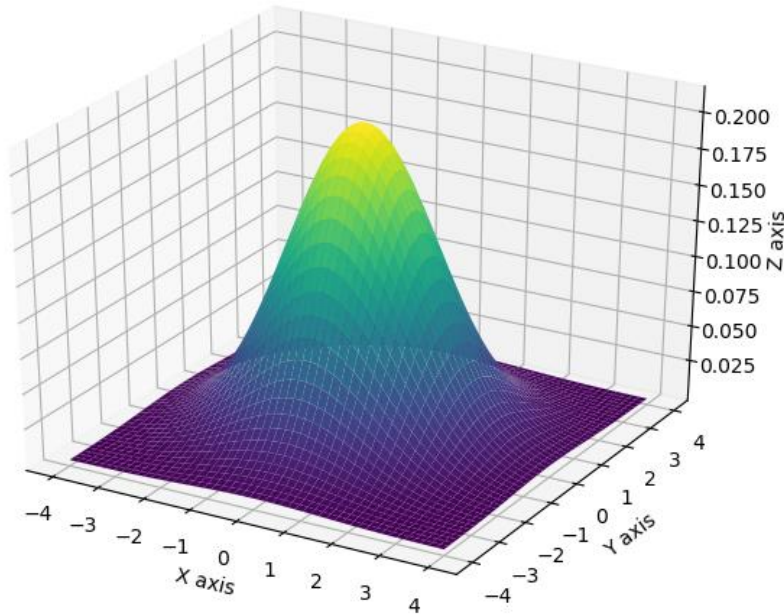
$$G(x, \sigma) = \frac{1}{\sqrt{2\sigma^2\pi}} e^{-\frac{x^2}{2\sigma^2}}$$



Effect of σ

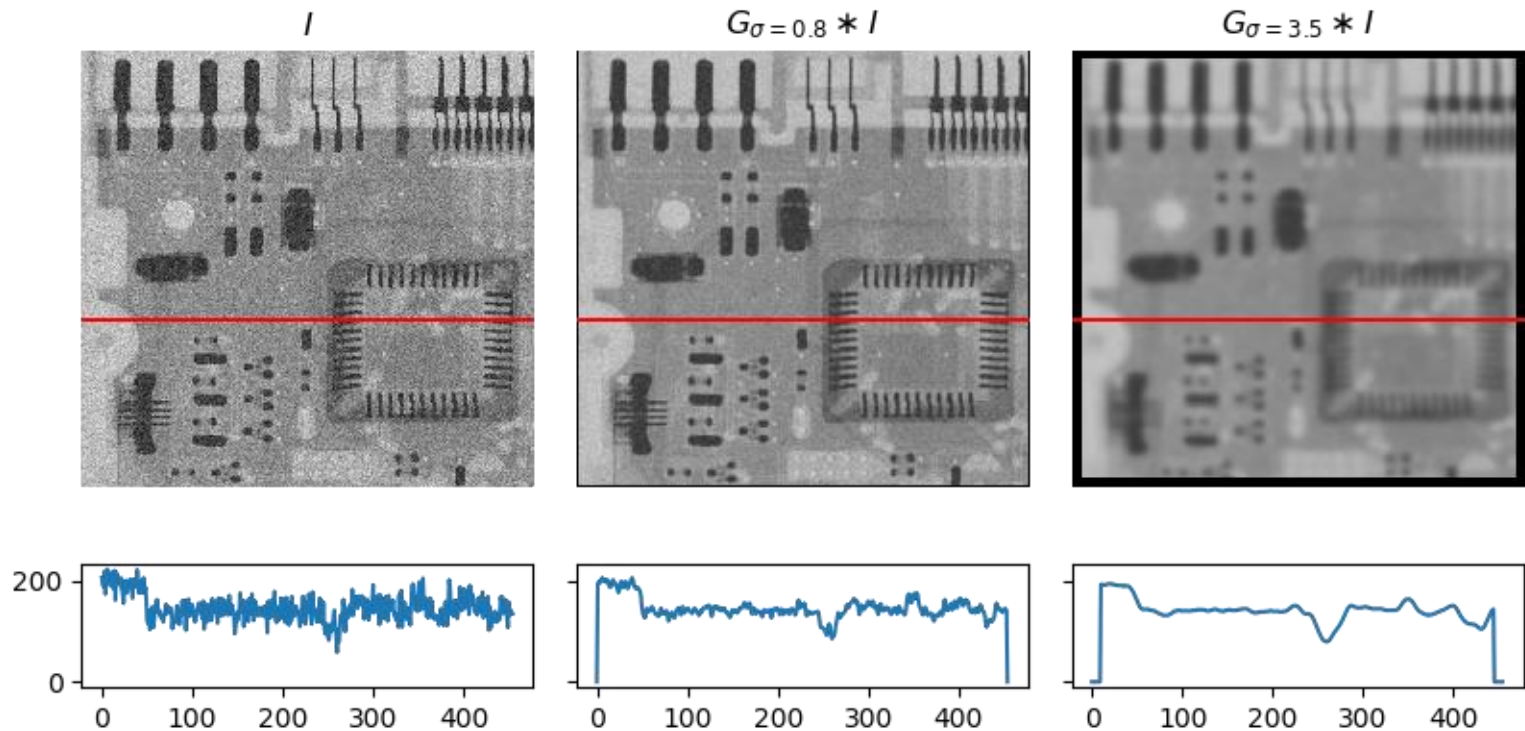


Gaussian Distribution – 2D



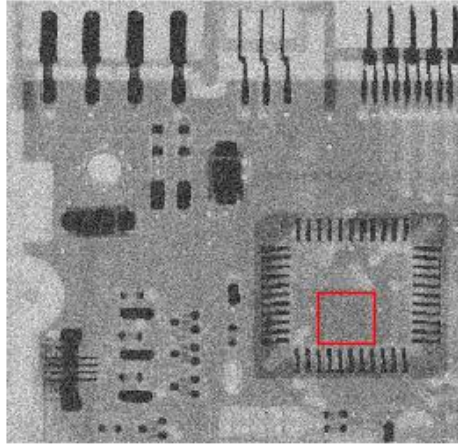
$$G(x, y, \sigma) = \frac{1}{2\sigma^2\pi} e^{-\frac{x^2+y^2}{2\sigma^2}}$$

“Looking” At The Intensity Profile

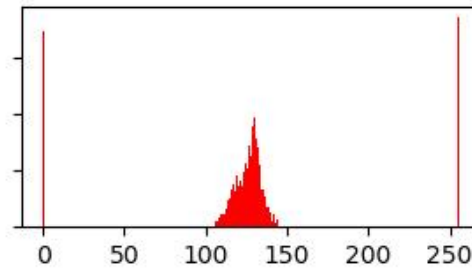
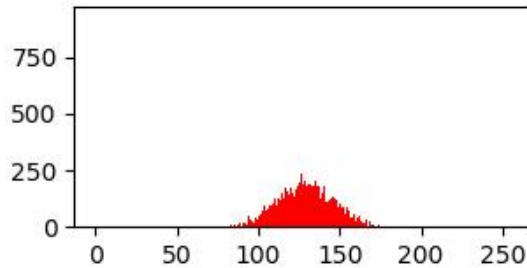
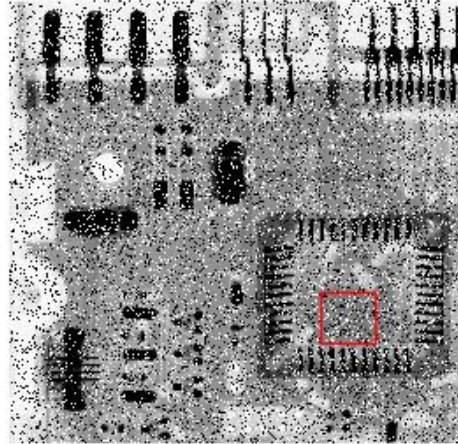


Gaussian Noise vs Salt & Pepper Noise

Gaussian noise

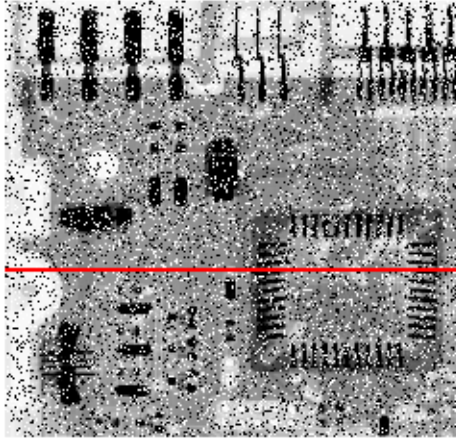


Salt & Pepper noise

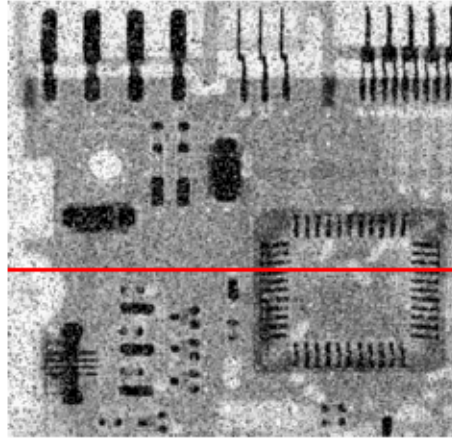


Filtering S&P: Attempt 1 (The Profile)

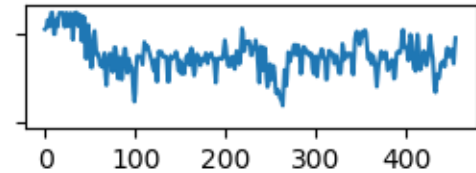
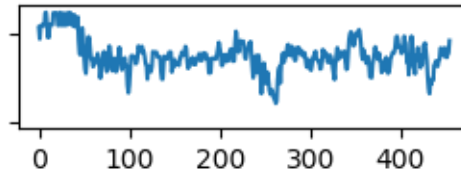
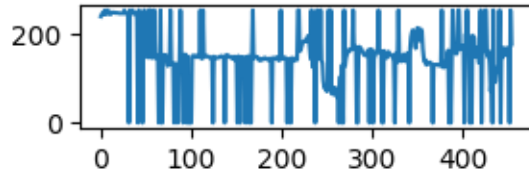
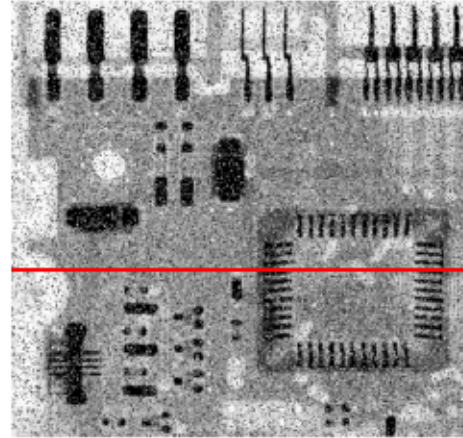
Noisy



Mean filtered

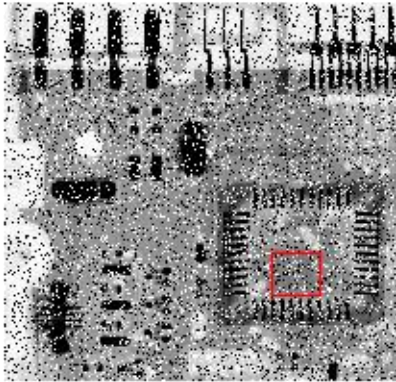


Gaussian filtered

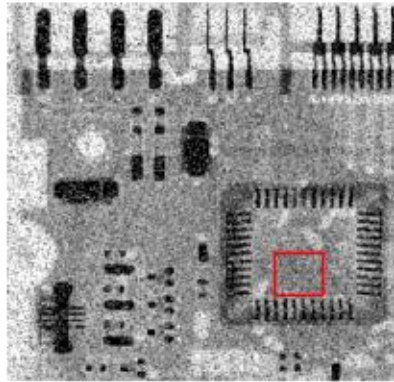


Filtering S&P: Attempt 1 (The Histogram)

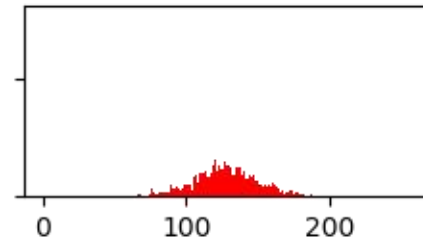
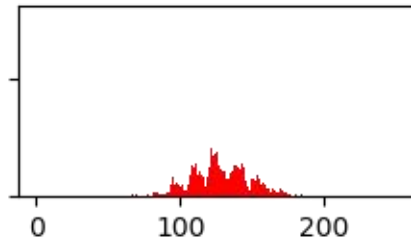
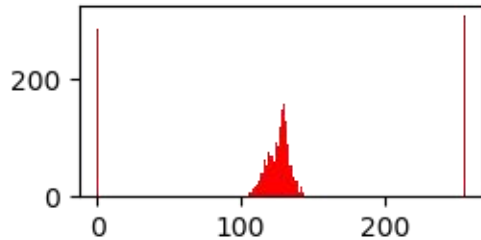
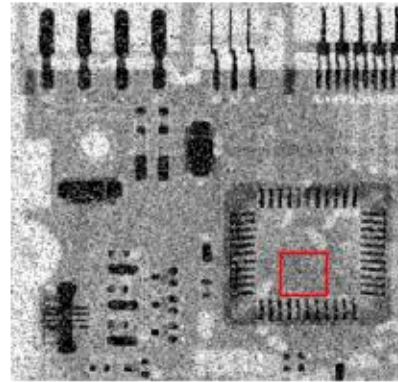
Noisy



Mean filtered



Gaussian filtered



Non-linear Filtering

- Does not rely on convolution/cross-correlation
- Rank filters are non-linear filters
- At each patch/neighbourhood, sort pixels by intensity and:
 - Min filter: take the first
 - Max filter: take the last
 - Median filter: take the central

3 x 3 Media Filtering (Example)

I'

	100	100	100	
	100	100	100	
	100	100	100	

median($\square$)

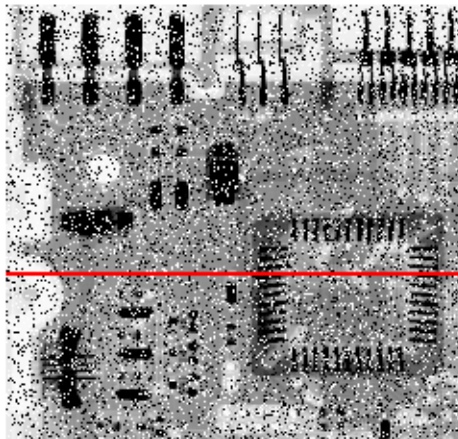
←.....

I

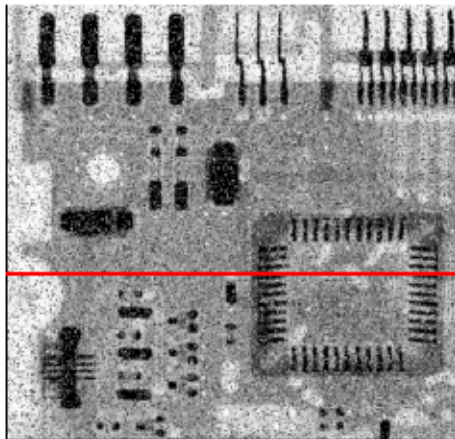
100	101	99	101	103
101	100	255	99	101
102	100	98	100	100
100	255	0	100	100
99	100	103	99	101

Filtering S&P: Attempt 2

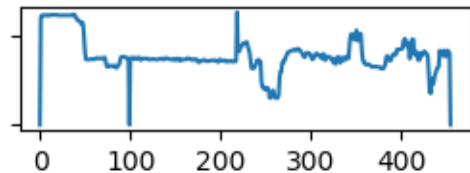
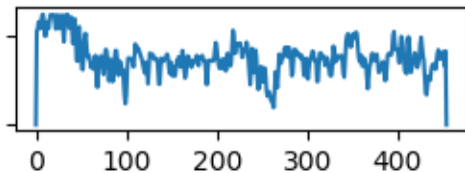
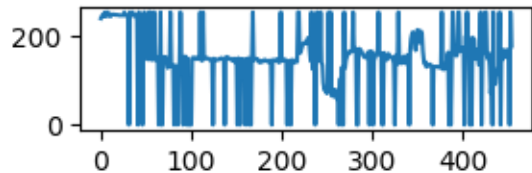
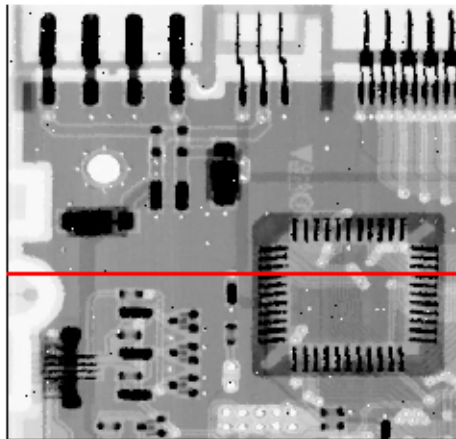
Noisy



Gaussian filtered



Median filtered

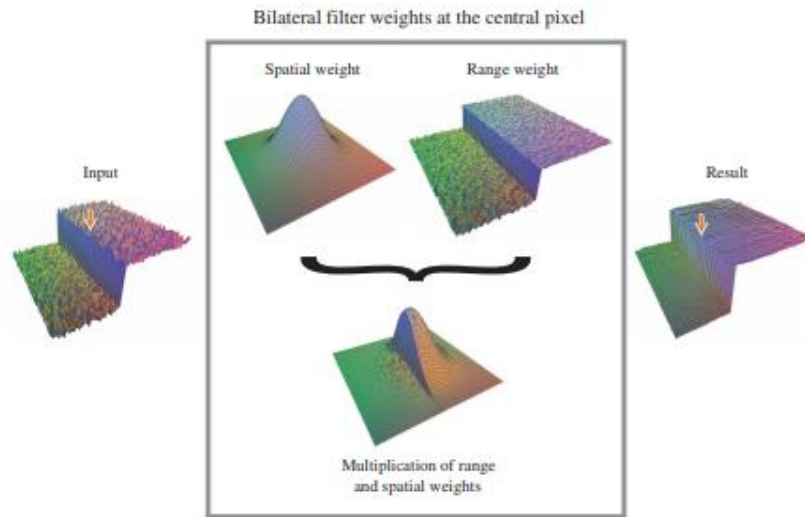


Bilateral Filter

- Mean and Gaussian filters smooth the edges (sometimes, not desired)
- Bilateral filter uses 2 criteria to smooth
 - Spatial distance from pixel to neighbours (G_s)
 - Intensity difference from pixel to neighbours (G_r)

$$BF[I]_{\mathbf{p}} = \frac{1}{W_{\mathbf{p}}} \sum_{\mathbf{q} \in \mathcal{S}} G_{\sigma_s}(\|\mathbf{p} - \mathbf{q}\|) G_{\sigma_r}(|I_{\mathbf{p}} - I_{\mathbf{q}}|) I_{\mathbf{q}},$$

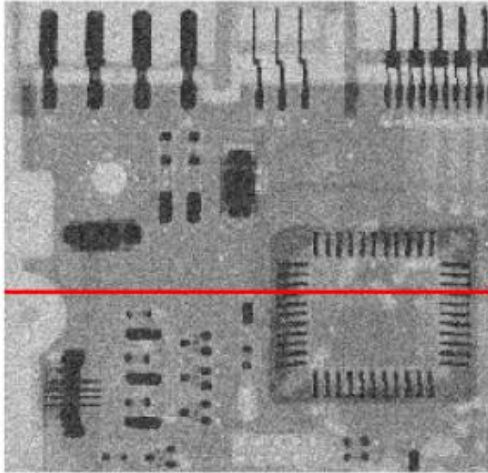
$$W_{\mathbf{p}} = \sum_{\mathbf{q} \in \mathcal{S}} G_{\sigma_s}(\|\mathbf{p} - \mathbf{q}\|) G_{\sigma_r}(|I_{\mathbf{p}} - I_{\mathbf{q}}|).$$



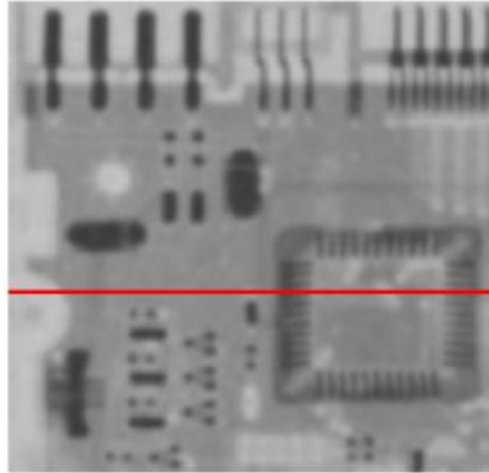
The figure is from Durand et al. "Fast bilateral filtering for the display of high dynamic-range images"

Bilateral Filter: Example

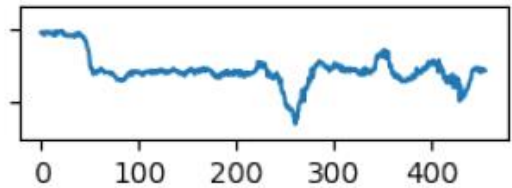
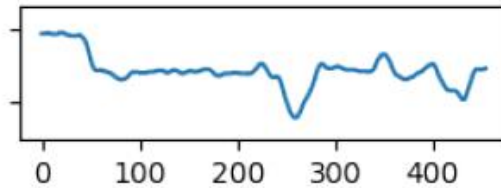
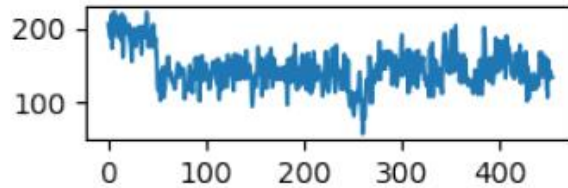
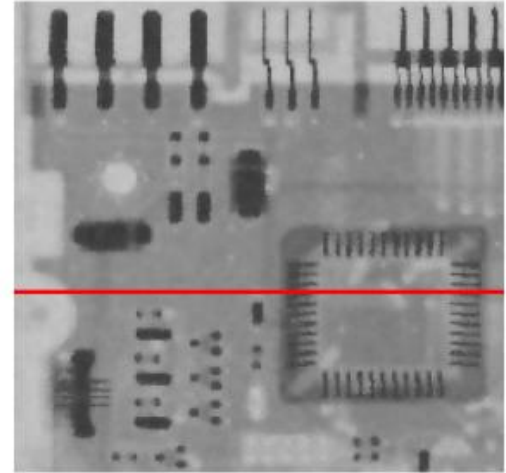
Noisy



Gaussian filtered



Bilateral filtered



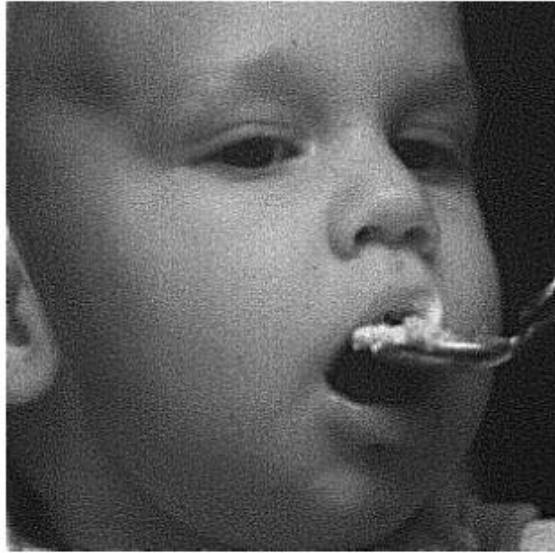
Review: Smoothing Filters

- Mean filter:
 - Quick to compute
 - Separable
 - Good for Gaussian-like noise
 - Blurs edges
- Gaussian filter:
 - Acceptably quick to compute
 - Separable
 - Good for Gaussian-like noise
 - Blurs edges
- Median filter:
 - Slow to compute
 - Good for impulse noise
 - Does not blur edges but not good for Gaussian-like noise
- Bilateral filter:
 - Slow to compute
 - Good for Gaussian-like noise
 - Does not blur edges

Sharpening Filters

- The opposite of smoothing
- Make images sharper
- AKA high-pass filters

Sharpen



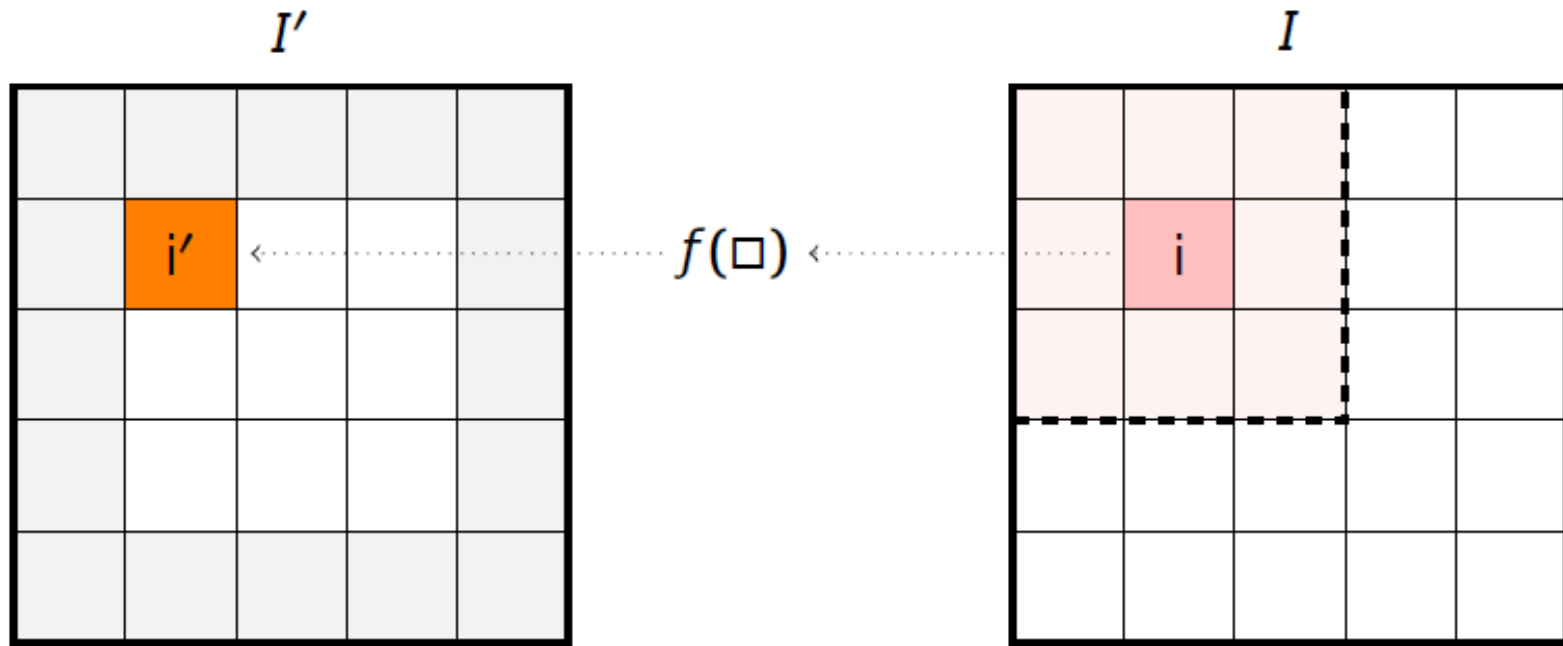
=

-1	-1	-1
-1	9	-1
-1	-1	-1

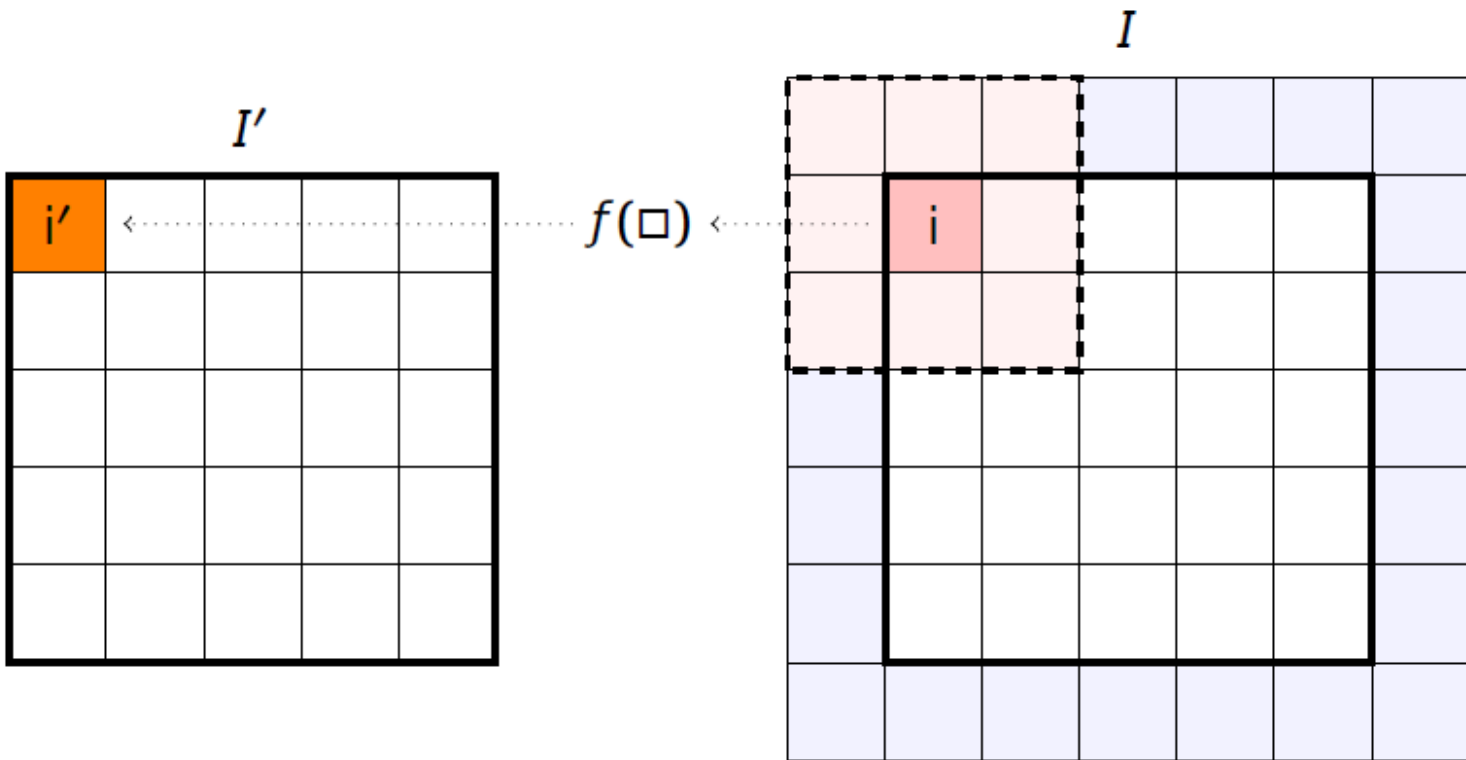
*



Dealing With Borders



If It Is on the Border? (Padding)



Padding: Constant

[illegible]

Padding: Replicate/Extend

0	0	0	1	2	3	4	4	4
0	0	0	1	2	3	4	4	4
0	0	0	1	2	3	4	4	4
5	5	5	6	7	8	9	9	9
10	10	10	11	12	13	14	14	14
15	15	15	16	17	18	19	19	19
20	20	20	21	22	23	24	24	24
20	20	20	21	22	23	24	24	24
20	20	20	21	22	23	24	24	24

Padding: Reflect/Mirror

6	5	5	6	7	8	9	9	8
1	0	0	1	2	3	4	4	3
1	0	0	1	2	3	4	4	3
6	5	5	6	7	8	9	9	8
11	10	10	11	12	13	14	14	13
16	15	15	16	17	18	19	19	18
21	20	20	21	22	23	24	24	23
21	20	20	21	22	23	24	24	23
16	15	15	16	17	18	19	18	18

Padding: Wrap

18	19	15	16	17	18	19	15	16
23	24	20	21	22	23	24	20	21
3	4	0	1	2	3	4	0	1
8	9	5	6	7	8	9	5	6
13	14	10	11	12	13	14	10	11
18	19	15	16	17	18	19	15	16
23	24	20	21	22	23	24	20	21
3	4	0	1	2	3	4	0	1
8	9	5	6	7	8	9	5	6

Review

- Non-linear filters are, usually, slower than linear
- Convolution with linearly separable filters is faster than with non-separable
- Mean and Gaussian filters are good for Gaussian noise
- Median filter is good for impulse noise
- Bilateral filter does not blur edges, however, it's computationally expensive
- Rule of thumb: Filter widths of 3 or 5 for real time applications
- Filter with widths > 5 are slow, OK for offline processing